

Round-14 Repaired Proof: Active Virtual Bags for the Patchwork-Bag Abstraction

GPT-5.5 Pro (OpenAI), repair after the G2/G3 round-13 failure

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Abstract

Round 13 repaired the round-12 shadow-amalgamation gap by forbidding vertical shadow entries and replacing them by co-bagged verticalization. The targeted G2 review found that this is not extractable at bounded width: the fixed satisfiable tower concept

$$\exists PP.T \sqcap \forall PP(\exists PP.T \sqcap (\forall PP.\exists PP.T \sqcap \forall DR.A))$$

forces unboundedly many remote obligation sources to be inserted into the same lower bag under literal D2. This document gives the repaired Theorem B proof. The repair abandons horizontal-only shadows. Universal sources are propagated as *virtual active ports*: their relation vectors may contain vertical atoms, but every finite active bag, including simultaneous shadows, must be a complete composition-closed RCC5 network and adjacent active bags must agree. Thus no remote source is inserted as a live port merely because its relation to a separator is vertical, while the old shadow-amalgamation counterexamples are excluded by finite active-network closure. The status remains the project status: strongly supported, not certified.

Contents

1	Patch statement	1
2	The active-virtual-bag discipline	2
3	Active shadow amalgamation	3
4	Coverage and truth	4
5	Extraction and width	5
6	Automaton delta	6
7	Machine checks	6
8	Replacement theorem statement	7

1 Patch statement

This is a replacement for round-13 Sections 3–8. The following clauses are withdrawn as proof devices:

- (i) D1's horizontal-only restriction on shadow rows;
- (ii) D2's verticalization fallback for universal shadows;
- (iii) the global default value DR for nonlocal undeclared pairs.

The retained ingredients are the round-12 bag/overlap infrastructure, strong EQ as identity, witness/request handling for existentials, the RCC5 patchwork theorem, and the normal-form Theorem A. Vertical propagation D3 is still a useful local saturation rule, but it is no longer the only way a vertical universal is protected in the truth lemma; exact active-shadow entries protect all universal roles uniformly.

Design principle. A universal source should not be converted into a port just because its current relation to a separator is PP or PPI. It should instead be kept as a virtual occurrence whose exact relation to each current port is declared. What made this unsafe in round 12 was not the presence of vertical entries by itself, but the absence of closure conditions between two simultaneously active shadows. The new discipline treats active shadows as virtual ports for the sole purpose of finite RCC5 closure and patchwork.

2 The active-virtual-bag discipline

Let T be a tree of finite bags. Each bag t has a finite port set P_t , a complete atomic RCC5 network N_t on P_t , and a type map $\tau_t : P_t \rightarrow \text{Typ}(C_0)$. Overlap maps identify manifestations of one global occurrence exactly as in round 12. Let V_T be the set of global occurrences. For an occurrence x , write $\text{tp}(x)$ for the common type of its manifestations, and put

$$\Omega_x = \{(R, D) : \forall R.D \in \text{tp}(x)\}.$$

Let

$$\text{Univ}_T = \{x \in V_T : \Omega_x \neq \emptyset\}$$

be the set of universal sources. These are the only occurrences that need long-range protection for the truth lemma.

Definition 2.1 (Active occurrence set). *For every bag t , define*

$$A_t = P_t \cup \text{Univ}_T.$$

If a universal source x is also represented by a live port in P_t , the live port and the virtual copy are the same occurrence. Thus A_t is a set of global occurrences, not a disjoint union of ports and shadows.

The set A_t may be infinite. Validity is therefore stated by finite-subset conditions. This is exactly the form needed by patchwork and compactness.

Definition 2.2 (AV-discipline). *An active-virtual-bag abstraction satisfies the inherited round-12 port, overlap, type, witness, request, and parity clauses, and additionally satisfies AV1–AV6 below.*

(AV1) **Finite active networks.** *For each bag t and every finite set $F \subseteq A_t$, the abstraction supplies a complete atomic network N_F^t on F such that:*

- (a) N_F^t restricts to the live bag network N_t on $F \cap P_t$;

- (b) if $F \subseteq F' \subseteq A_t$ are finite, then $N_{F'}^t \upharpoonright F = N_F^t$;
- (c) N_F^t satisfies the RCC5 composition table on every triple;
- (d) EQ occurs only on the diagonal, i.e. distinct occurrences are never related by EQ.

Equivalently, one may view AV1 as a coherent complete atomic relation N_t^* on the active set A_t , with every finite restriction closed.

- (AV2) **Exact active rows.** If $x \in \text{Univ}_T$ and $p \in P_t$, then $N_t^*(x, p)$ is the declared relation from the universal source x to the current port p . The value may be DR, PO, PP, or PPI; it is EQ only when p is a manifestation of x itself. There is no horizontal-only restriction and no verticalization fallback.
- (AV3) **Active pair coherence.** If s, t are adjacent bags, then N_s^* and N_t^* agree on every finite subset of $A_s \cap A_t$. In particular, a pair of universal sources has one relation wherever they are simultaneously active, and a universal-source/continued-port row value is rigid across the overlap.
- (AV4) **Universal type check.** For every bag t , every universal source $x \in \text{Univ}_T$, every live port $p \in P_t$, and every $\forall R.D \in \text{tp}(x)$,

$$N_t^*(x, p) = R \implies D \in \text{tp}(p).$$

The same-bag check is the special case $x \in P_t$. Strong-EQ formulas, if not normalized away, are checked only on the identical occurrence.

- (AV5) **Existential witnesses and vertical requests.** The round-12 and round-13 witness clauses are retained: every $\exists R.D \in \text{tp}(p)$ has a local, saturated side-context, or parity-fulfilled vertical request witness with exact bag relation R and target type containing D . Request-source liveness is used for existential fulfillment, not for universal-shadow verticalization.
- (AV6) **No default for inactive pairs.** If a pair of occurrences is not co-bagged and neither occurrence is in Univ_T , no relation is preassigned to that pair. It is completed only by the patchwork/compactness construction. There is therefore no global DR default.

Remark 2.3 (How this differs from round 13). Round 13 forced vertical source–port entries to become live co-baggings. AV2 keeps them virtual. Round 13 tried to recover patchwork by making every shadow-incident side horizontal. AV1 instead demands the actual missing condition: every finite set of live ports and active shadows already forms a closed RCC5 network. This is the extended-bag repair route identified by WP15, now used in place of the horizontal-row route.

3 Active shadow amalgamation

Lemma 3.1 (Active family is tree-shaped). *For each occurrence u , the set of bags whose active set contains u is connected. Adjacent active networks agree on their common occurrences.*

Proof. If $u \in \text{Univ}_T$, then $u \in A_t$ for every bag t , so its active trace is the whole tree. If $u \notin \text{Univ}_T$, then u appears only as a live port, and its live trace is connected by the inherited overlap/running-intersection condition. Agreement of adjacent active networks is AV3; on live ports this is the ordinary overlap agreement, and on pairs involving universal sources it is active pair coherence. $\square$

Lemma 3.2 (Finite active patchwork). *Let Q be a finite connected subtree of bags and let $F \subseteq V_T$ be finite. There is a complete atomic RCC5 network on*

$$F_Q = F \cup \bigcup_{t \in Q} P_t$$

that extends every live bag value in Q and every active value $N_t^(x, p)$ or $N_t^*(x, y)$ with $t \in Q$ and $x, y \in F_Q \cap A_t$.*

Proof. For each $t \in Q$, let

$$B_t = F_Q \cap A_t.$$

The set B_t is finite because Q and all live bags are finite and F is finite. By AV1, $N_t^* \upharpoonright B_t$ is a complete atomic composition-closed RCC5 network. By Lemma 3.1, adjacent bags agree on the overlap $B_s \cap B_t$. The family $(B_t, N_t^* \upharpoonright B_t)_{t \in Q}$ is therefore a finite tree-shaped family of complete atomic closed networks agreeing on overlaps. The round-12 finite tree-patching lemma gives a complete atomic RCC5 network on the union F_Q extending all these values. $\square$

Theorem 3.3 (Active shadow amalgamation). *Every valid active-virtual-bag abstraction has a total abstract complete atomic RCC5 frame ρ_T on V_T extending all live bag values and all active virtual-bag values.*

Proof. For every finite $F \subseteq V_T$, choose a finite connected subtree Q_F containing at least one live manifestation of each occurrence of $F \setminus \text{Univ}_T$. Occurrences in $F \cap \text{Univ}_T$ are active in every bag, so no special bag is needed for them. By Lemma 3.2, there is a complete atomic RCC5 network on a finite superset of F extending all active and live constraints encountered in Q_F ; restrict it to F . Thus every finite set of occurrences has a closed atomic labelling compatible with the declared data.

Now use the same compactness argument as round 12, with the F5 wording repair “some completion” rather than “the completion”. The product space $\text{Rel}^{V_T^2}$ is compact. For a finite set F , let C_F be the closed set of total labellings whose restriction to F is extendable to one of the finite completions above and which respect converse symmetry and strong identity on F . The preceding paragraph gives the finite intersection property: a larger finite subtree and a larger finite occurrence set witness any finite family of requirements. A point in the intersection of all C_F ’s is a total labelling ρ_T . Every RCC5 composition axiom is a condition on a finite triple, hence holds in ρ_T . Since all finite completions are atomic and use EQ only for identical occurrences, the same is true of ρ_T . $\square$

Remark 3.4 (Why WP15/WP20 cannot recur). *The old counterexamples put two shadows in a bag with independently legal one-point rows but no declared relation between the shadows. AV1 has no such case: the two shadows and any live port form a finite active network, and every triangle among them must satisfy the RCC5 table. For example, $xPPy, yDRp$, and $xPOp$ is rejected because $\text{Comp}(PP, DR) = \{DR\}$. This is exactly the WP20 finite check.*

4 Coverage and truth

The old round-13 Coverage Lemma had a difficult walk induction with three cases: co-bagged, row-covered, or uniformly vertical. Under active virtual bags, coverage is simpler and relation-exact.

Lemma 4.1 (Active coverage). *Let T be a valid active-virtual-bag abstraction and let ρ_T be supplied by Theorem 3.3. If $x \in \text{Univ}_T$ and $y \in V_T$, then for every bag t containing a live manifestation of y , the active value $N_t^*(x, y)$ is defined and*

$$\rho_T(x, y) = N_t^*(x, y).$$

Consequently, if $\forall R.D \in \text{tp}(x)$ and $\rho_T(x, y) = R$, then $D \in \text{tp}(y)$.

Proof. Since $x \in \text{Univ}_T$, we have $x \in A_t$ for every bag t . If t contains a live manifestation of y , then $y \in P_t \subseteq A_t$, so $N_t^*(x, y)$ is one of the active values. Theorem 3.3 extends all active values, hence $\rho_T(x, y) = N_t^*(x, y)$. If this value is R and $\forall R.D \in \text{tp}(x)$, AV4 gives $D \in \text{tp}(y)$. $\square$

Remark 4.2 (Mixed walks). *There is no residual mixed-walk case. A path from a source manifestation to a target bag may pass through any mixture of horizontal and vertical separators, but the source remains a virtual active occurrence at the target bag. The truth proof uses the endpoint active value, not a composition of the walk. D3 may still propagate vertical universal formulas along selected uniform support chains, but the truth lemma no longer depends on every vertical endpoint being reached by such a chain.*

Theorem 4.3 (Truth lemma, AV form). *Let T be a valid active-virtual-bag abstraction with a root port whose type contains C_0 , and let ρ_T be the frame of Theorem 3.3. Interpret concept names by types: $x \in A^{T_x}$ iff $A \in \text{tp}(x)$. Then for every occurrence x and every $C \in \text{Cl}(C_0)$,*

$$C \in \text{tp}(x) \iff \mathcal{I}_T, x \models C.$$

Proof. By induction on C . Atomic and Boolean cases are immediate from the type conditions.

For $C = \exists R.D$, if $\exists R.D \in \text{tp}(x)$, AV5 supplies a witness: either a local or saturated side-context port, or a parity-fulfilled vertical request target. In all cases the relevant relation is a live bag value, hence is extended by ρ_T , and the target type contains D . The induction hypothesis gives a semantic witness. Conversely, if $\exists R.D \notin \text{tp}(x)$, type completeness gives $\forall R.\bar{D} \in \text{tp}(x)$. The universal case below implies that no R -successor satisfies D .

For $C = \forall R.D$, assume $\forall R.D \in \text{tp}(x)$ and let y be any occurrence with $\rho_T(x, y) = R$. Then $x \in \text{Univ}_T$, so by Lemma 4.1, $D \in \text{tp}(y)$. The induction hypothesis gives $\mathcal{I}_T, y \models D$. Since y was arbitrary, the universal holds. Conversely, if $\forall R.D \notin \text{tp}(x)$, then $\exists R.\bar{D} \in \text{tp}(x)$, and the existential case supplies a counter-witness to the universal. $\square$

Corollary 4.4 (Realization direction of Theorem B, AV form). *Every valid active-virtual-bag abstraction with a root port containing C_0 unfolds to a model of C_0 over an abstract complete atomic RCC5 frame with strong equality.*

5 Extraction and width

Theorem 5.1 (Extraction direction of Theorem B, AV form). *If C_0 is satisfiable, then some saturated split-forest presentation of C_0 of width at most the Theorem A bound $K_A(C_0)$ admits a valid active-virtual-bag abstraction. No extra port manifestations are introduced for universal-shadow propagation.*

Proof. Let $\mathcal{M}$ be a satisfying abstract complete atomic RCC5 model, and let Φ be the witness-generated saturated split-forest presentation obtained from Theorem A after thinning. Its live bags have width at most $K_A(C_0)$. Use exactly these live bags, overlap maps, bag networks, type labels, local witnesses, side contexts, and vertical existential requests.

For every occurrence x , set $\text{tp}(x) = \{C \in \text{Cl}(C_0) : \mathcal{M}, x \models C\}$. For every bag t , put $A_t = P_t \cup \text{Univ}_T$. For each finite $F \subseteq A_t$, define N_F^t to be the restriction of the true relation of $\mathcal{M}$ to F . Because $\mathcal{M}$ is a complete atomic RCC5 frame with strong equality, every such finite active network is complete, atomic, composition-closed, and has EQ only on identical occurrences. The restriction to live ports is exactly the bag network of Φ . If two finite active sets are nested, their true-relation

restrictions are compatible. Adjacent active networks agree because they are restrictions of the same global relation. Thus AV1–AV3 hold.

AV4 holds because the types are read from $\mathcal{M}$: if $\forall R.D \in \text{tp}(x)$ and the true relation from x to a live port p is R , then $\mathcal{M}, p \models D$, hence $D \in \text{tp}(p)$. AV5 is inherited from the selected witnesses and parity request paths in the saturated presentation. AV6 is a convention on what is not declared. Therefore the extracted object is a valid AV-abstraction.

The live port sets are unchanged from Φ . Active shadows are virtual relations used by the verifier and patchwork proof; they are not manifestations and do not increase bag width. Hence the width remains at most $K_A(C_0)$. $\square$

Corollary 5.2 (The G2a tower is absorbed). *For the fixed G2a witness concept, a finite prefix $a_0\text{PP}a_1\text{PP}\cdots\text{PP}a_m$ may have all remote ancestors a_k active at the root edge bag. The relation from a_k to a lower separator is PPI, but this is an AV2 virtual entry, not a reason to insert a_k as a live port. The root edge bag keeps its two live ports while the active network records arbitrarily many virtual vertical entries. Thus the round-13 unbounded manifestation count disappears.*

6 Automaton delta

This section records the finite-state implementation obligation for Theorem C. It is included so that the proof document is internally consistent, but it should be treated as a review target rather than a mechanized certification.

The live input alphabet is no larger than in round 12, apart from replacing the old horizontal-row/verticalization annotations by active-shadow monitor annotations. Shadows are not listed as an unbounded part of a bag label; as in round 12, they are generated by alternating run copies.

Single-shadow states.

A state $q_{\text{sh}}(\sigma)$ carries a finite shadow profile $\sigma = (\tau, \Omega, \vec{r})$, where $\tau \in \text{Typ}(C_0)$, $\Omega \subseteq \text{Rel} \times \text{Cl}(C_0)$, and $\vec{r} \in \text{Rel}^{P_t}$ is the relation vector to the current finite port list. Unlike round 13, $\vec{r}$ may contain PP and PPI. The transition checks AV4 for all entries of $\vec{r}$ and propagates the profile to adjacent bags with overlap rigidity.

Pair states.

A state $q_{\text{pair}}(\sigma_1, e, \sigma_2)$, with $e \in \text{Rel}$, carries the mutual relation between two active shadows. At a bag it checks every triangle formed by the two shadows and one live port. Along an overlap it keeps e rigid. If one shadow later becomes live, the state checks that e agrees with the live bag value.

Triple states.

A state $q_{\text{tri}}(\sigma_1, e_{12}, \sigma_2, e_{23}, \sigma_3, e_{13})$ checks $e_{13} \in \text{Comp}(e_{12}, e_{23})$ and the two cyclic companion constraints. Together with the pair states, this is the finite-control form of AV1 for active-shadow triples.

No default state.

There is no state assigning DR to untracked nonlocal pairs. Nonactive pairs are left for patchwork completion.

The number of possible profiles is finite because $|P_t| \leq K_A(C_0)$, $\text{Typ}(C_0)$ is finite, and Rel is finite. The tuple monitors therefore add finitely many states. The parity condition for vertical existential requests is unchanged. The usual two-way alternating parity emptiness theorem then applies to the repaired language.

Remark 6.1 (Cold-review target). *The semantic proof above does not depend on a particular encoding of the tuple monitors. A mechanized round should still spell out the spawning discipline for pair and triple states, especially for two sources first introduced in disjoint subtrees. The finite data carried by such states are clear; the review target is the bookkeeping, not an additional RCC5 fact.*

7 Machine checks

The new script

`verification/wp25_active_virtual_bag_repair.py`

checks the semantic finite claims introduced by this repair. The accompanying WP21–WP24 scripts check profile-copy expansion, the G2a profile-count regression, active-closure regressions, and vertical-walk facts retained from the G3 review. Its output is:

```
WP25 -- active-virtual-bag repair checks
=====
A. robust G2a witness SAT by cover-tree oracle: PASS closure=18
B. G2a prefix width: old fallback unbounded, new live ports bounded: PASS
  m chain_closed old_root_ports new_root_ports virtual_shadows
  1         True           2           2           0
  2         True           3           2           1
  3         True           4           2           2
  4         True           5           2           3
  5         True           6           2           4
  6         True           7           2           5
  7         True           8           2           6
  8         True           9           2           7
  9         True          10           2           8
 10         True          11           2           9
C. WP20 bad shadow-shadow triangle rejected by active closure: PASS
  closure conflict: ('composition', 'x', 'y', 'p', 'PO', 'PP', 'DR', ['DR'])
D. random concrete active restrictions closed: PASS
=====
WP25 OVERALL: PASS
```

WP14 remains the check for the finite RCC5 patchwork theorem, WP15 identifies the original shadow-amalgamation failure and its extended-bag repair route, WP17 remains useful for the horizontal-row facts but is no longer load-bearing for D1, and WP18–WP20 are regression checks for the G2/G3 failures that this round removes.

8 Replacement theorem statement

Combining Theorem A, Theorem 5.1, Theorem 3.3, Theorem 4.3, and the automaton delta yields the repaired finite-abstraction theorem:

Theorem 8.1 (Theorem B, round-14 AV form). *For every input concept C_0 , saturated split-forest presentations of width at most $K_A(C_0)$ are soundly and completely represented, up to $\text{Cl}(C_0)$ -types, by active-virtual-bag abstractions. A valid abstraction with a root port containing C_0 realizes to a model of C_0 ; every satisfying model has such an abstraction extracted from a Theorem A presentation without increasing live bag width.*

Corollary 8.2 (Decidability status). *Together with the finite two-way alternating parity automaton for the repaired language and the standard emptiness decision, concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ over abstract complete atomic RCC5 frames with strong equality is decidable. In project terminology, this is strongly supported, not certified, until a cold review or mechanization finds no defect.*

References

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